

Observed Pairwise Data

Testing-time difference: t_{ij}

Consensus genetic distance: g_{ij}

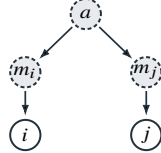
Unobserved Transmission Scenarios (S_M)

Ancestor-Descendant $H_{AD}(m)$



Direct transmission $i \rightarrow j, m = 0$

Common Ancestor $H_{CA}(m_i, m_j)$



Co-primary infection $(m_i, m_j) = (0, 0)$

Process-Based Model

Gamma stages ($E/P/I$)

Incubation period: $\tau_{inc} = y_E + y_P$

Generation time: $\tau_{gen} = y_E + y_{toit}$

Observation interval: $\tau_{obs} = \tau_{inc} + x_{test}$

Clock rate: $\lambda = \nu L / 365$ (strict or relaxed)

Monte Carlo Time Draws

AD: $T_s = \sum_{r=0}^m \tau_{gen,r} + \tau_{obs,j} - \tau_{obs,i}$

CA: $T_s = \sum_{s=1}^{m_j+1} \tau_{gen,s}^{(j)} - \sum_{r=1}^{m_i+1} \tau_{gen,r}^{(i)} + \tau_{obs,j} - \tau_{obs,i}$

Signed difference for temporal ordering

Monte Carlo Genetic Draws

AD: $B_s = \max\{0, T_s\}$

CA: $B_s = \sum \tau_{gen}^{(i)} + \sum \tau_{gen}^{(j)} + \tau_{obs,i} + \tau_{obs,j}$

Deterministic: $G_s = \lambda B_s$

Stochastic: $G_s \sim \text{Poisson}(\lambda B_s)$

Temporal Compatibility

$$C_{T,s} = 1 - 2|\Pr(T_s \leq t_{ij}) - 0.5|$$

Genetic Compatibility

$$C_{G,s} = 1 - 2|\Pr(G_s \leq g_{ij}) - 0.5|$$

Aggregated Compatibility Score

Scenario: $C_s(i, j) = C_{T,s} \cdot C_{G,s}$

Target Set: $C_{S_\star}(i, j) = \sum_{s \in S_\star} C_s(i, j)$